Euclid's Elements — Book II
Proposition 10

Formal Reconstruction — Technical Documentation

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CGJteamLab

If a straight line is bisected, and a straight line is added to it in a straight line, then the square on the whole with the added straight line and the square on the added straight line both together are double the sum of the square on the half and the square described on the straight line made up of the half and the added straight line as on one straight line.

1 Introduction

Proposition II.10 is the external-section companion to Proposition II.9, but the current Lean development does not prove it by invoking II.9. Instead it reconstructs Euclid's second source configuration directly. The midpoint C lies between A and B, while the new point D lies beyond B:

$$A - C - B - D.$$

Euclid then erects the same perpendicular at C, constructs a parallel parallelogram, produces two lines to a new intersection G, proves two isosceles equalities, and uses Proposition I.47 four times.

The public theorem states that the two squares on AD and DB have the same content as two copies of the square on AC together with two copies of the square on CD. A useful modern mnemonic is

$$AD^2 + DB^2 = 2(AC^2 + CD^2).$$

If $AC = CB = a$ and $BD = x$, then $AD = 2a + x$ and $CD = a + x$, so the familiar algebraic shadow is

$$(2a + x)^2 + x^2 = 2a^2 + 2(a + x)^2.$$

Neither formula is the formal object proved by Lean. The theorem constructs four actual square representatives and relates their formal sums by `HilbertScissorsEquicomplementable`. It introduces neither numerical area nor multiplication of segment lengths.

The production source begins with

```
import CGJteamLab.Proposition2_9Construction
import CGJteamLab.Proposition34
import CGJteamLab.HilbertAngleDecomposition
import CGJteamLab.Proposition06
import CGJteamLab.Proposition32
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
import CGJteamLab.HilbertSquareTransport
```

The import of `Proposition2_9Construction` is an infrastructure reuse, not a proof by Proposition II.9. II.10 does not import `Proposition2_9.lean` and does not call `euclid_proposition_2_9_oriented`. It reuses the genuinely generic perpendicular-and-copy helper extracted while II.9 was being developed.

2 Euclid's Statement and Classical Configuration

2.1 The statement

Euclid's Proposition II.10 begins with a line AB bisected at C and a segment BD added in the same straight line beyond B. In the lettering used here the order is therefore

$$A - C - B - D.$$

Euclid's claim is

$$\mathrm{Sq}(AD) + \mathrm{Sq}(DB) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD),$$

where the equality is an equality of geometric content between plane figures [?].

2.2 The classical construction

At C erect CE perpendicular to AB and choose it so that

$$CE = AC = CB.$$

Join EA and EB. Through E draw EF parallel to AD, and through D draw DF parallel to CE. The quadrilateral $CDFE$ is therefore a parallelogram [?]. This is the first complete auxiliary state of the construction.

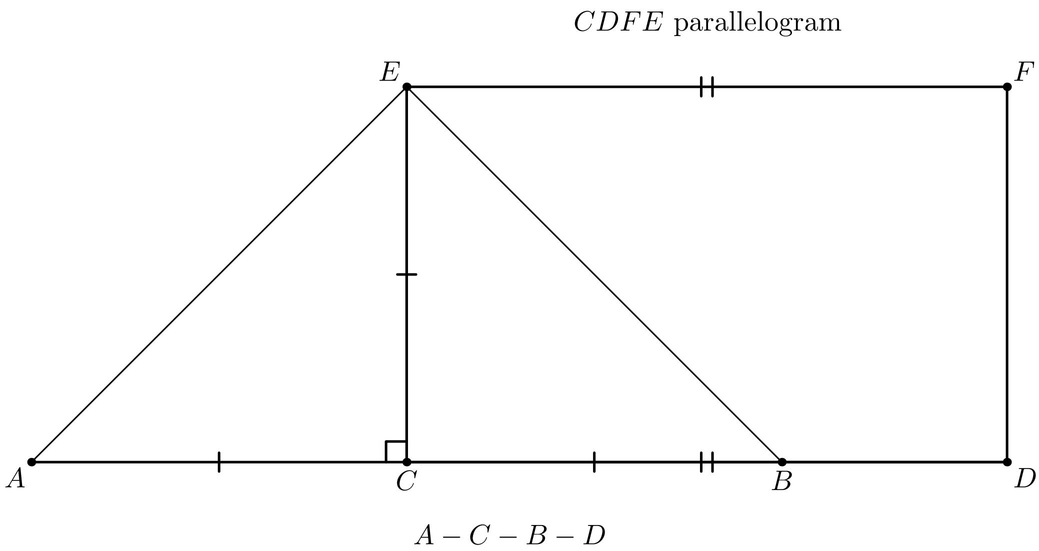

Figure 1: The first geometric state of II.10. The baseline has order $A - C - B - D$; the copied perpendicular satisfies $CE \cong AC \cong CB$; and F closes the auxiliary parallelogram $CDFE$. The joins EA and EB are already present. The point G and all metric conclusions depending on it are deliberately absent because they have not yet been constructed.

The next state appears when EB is produced beyond B and FD is produced beyond D until the two carriers meet at G; then AG is joined [**?**]. The strict positions are

$$E - B - G, \qquad F - D - G.$$

They matter formally. The first controls the extension of the right angle AEB to AEG and the vertical-angle argument at B; the second controls the right-angle and same-ray transports at D and F.

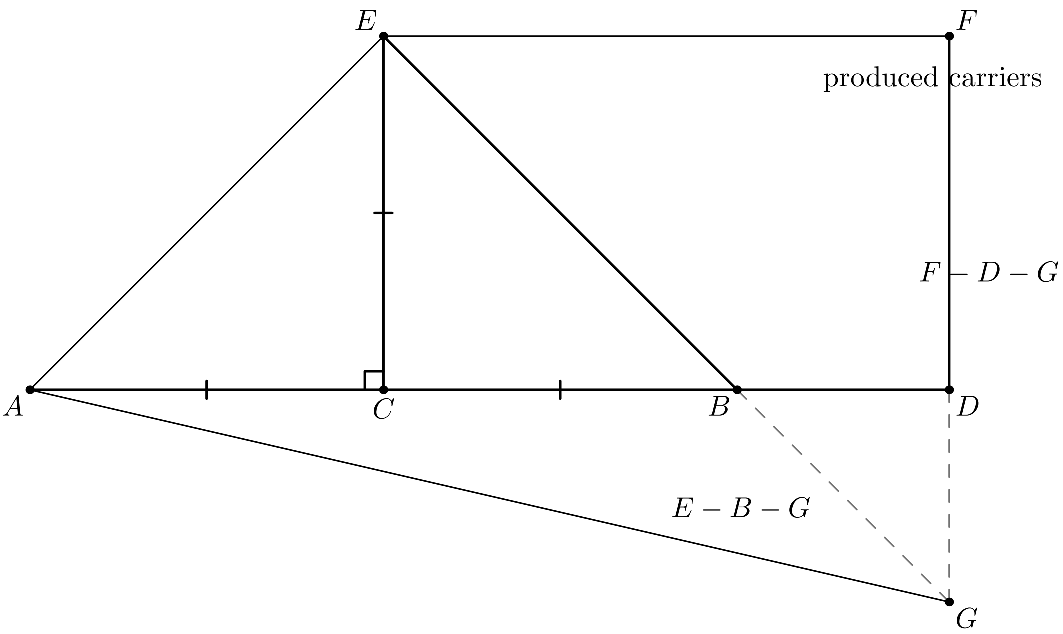

Figure 2: The second geometric state adds the intersection G. The dashed terminal portions of the two produced carriers emphasize that G lies beyond B on the carrier EB and beyond D on the carrier FD, giving exactly $E - B - G$ and $F - D - G$. The later conclusions $DB \cong DG$ and $GF \cong EF$ are not yet marked: their proof requires the subsequent angle blocks.

Diagrammatic audit. The original single diagram displayed the entire finished construction at once, including equalities that are proved only later. The audited sequence separates the actual proof states: first the perpendicular copy and parallelogram, then the oriented intersection, then the angle consequences, and finally the four Pythagorean triangles. Pure representative transport in the scissors layer receives no additional picture because it creates no new geometric configuration.

3 Classical Proof

The proof first turns the construction into two metric bridges and then compares two decompositions of the same square on AG.

Because $AC = CE$, Proposition I.5 gives

$$\angle EAC = \angle AEC.$$

Since $\angle ACE$ is right, Proposition I.32 makes each of these angles half a right angle. The same argument in the isosceles triangle CEB, using $CE = CB$, shows that $\angle CEB$ and $\angle EBC$ are also half right angles. Therefore

$$\angle AEB$$

is right.

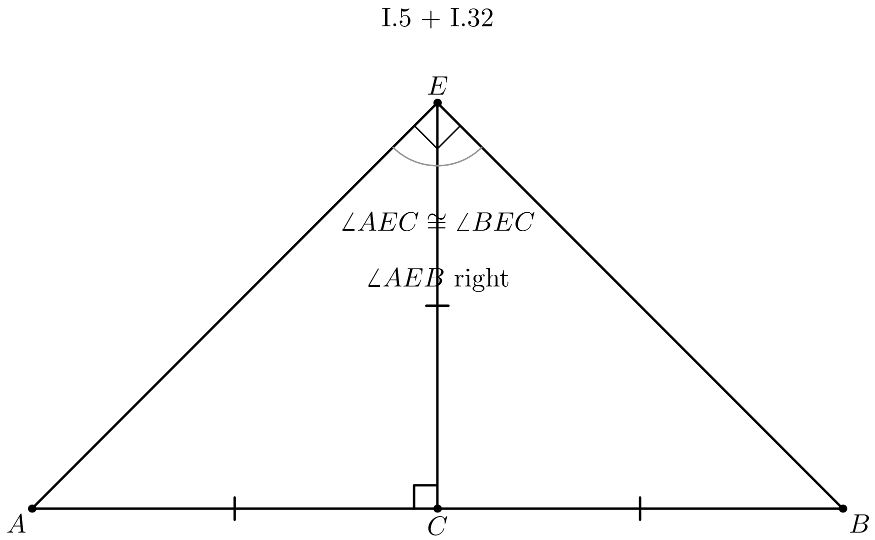

Figure 3: The two isosceles triangles ACE and CEB share the copied perpendicular CE. The equal segments $AC \cong CE \cong CB$ give the paired base angles by I.5; I.32 then organizes them as equal half-right components at E. Their union is the right angle AEB. The formal interior-ray witnesses R and S are omitted because their role is to certify this decomposition, not to introduce a new visible vertex of the classical configuration.

Since EB and FD have been produced to G, the vertical-angle theorem I.15 identifies the half-right angle at B with $\angle DBG$. The angle $\angle BDG$ is right by the parallel construction and I.29. Proposition I.32 then shows that the remaining angle $\angle DGB$ is half right. Hence

$$\angle DBG = \angle DGB,$$

and Proposition I.6 gives

$$DB = DG.$$

For the second isosceles bridge, the angle $\angle EGF$ is half right. The angle at F is right because it corresponds to the right angle at C in the parallelogram/parallel configuration. Proposition I.32 therefore makes $\angle FEG$ half right as well, and I.6 gives

$$GF = EF.$$

The opposite sides of parallelogram $CDFE$ are equal by I.34, so

$$EF = CD.$$

Consequently

$$GF = EF = CD.$$

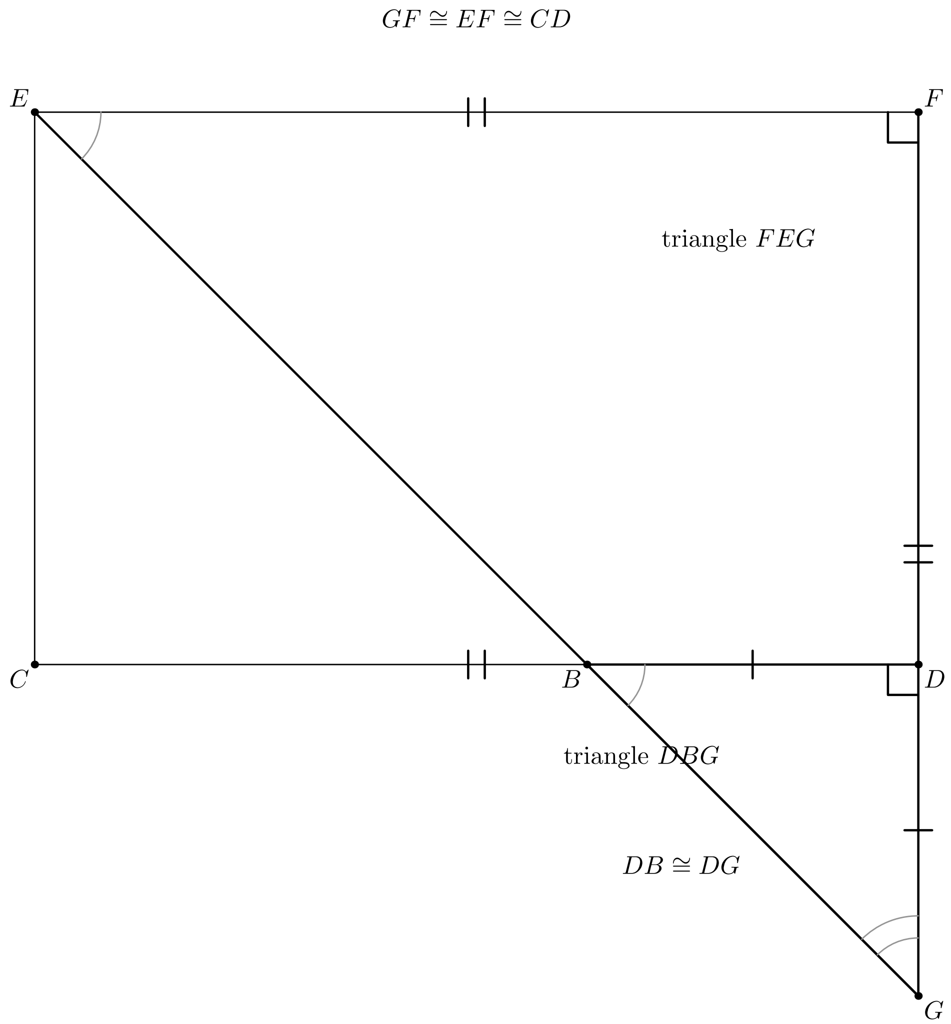

Figure 4: Once the orders $E - B - G$ and $F - D - G$ are known, the source angle argument acts on two visible triangles. In DBG, the vertical-angle and right-angle chain makes the angles at B and G congruent, so I.6 yields $DB \cong DG$. In FEG, the corresponding half-right-angle argument and the right angle at F give $GF \cong EF$; I.34 then transports the latter length to CD. Local angle-decomposition witnesses are omitted.

Now apply I.47 four times. In the right triangle ACE,

$$\mathrm{Sq}(AE) = \mathrm{Sq}(AC) + \mathrm{Sq}(CE) = 2\,\mathrm{Sq}(AC).$$

In the right triangle FEG,

$$\mathrm{Sq}(EG) = \mathrm{Sq}(FE) + \mathrm{Sq}(FG) = 2\,\mathrm{Sq}(CD).$$

Because $E - B - G$, the right angle AEB extends along the same ray to give $\angle AEG$ right. Thus I.47 in triangle EAG yields

$$\mathrm{Sq}(AG) = \mathrm{Sq}(AE) + \mathrm{Sq}(EG) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD).$$

At D, the parallel/right-angle construction and the collinear order on the base give $\angle ADG$ right. Hence I.47 in triangle DAG gives

$$\mathrm{Sq}(AG) = \mathrm{Sq}(AD) + \mathrm{Sq}(DG) = \mathrm{Sq}(AD) + \mathrm{Sq}(DB).$$

Comparing these two descriptions of the square on AG proves II.10.

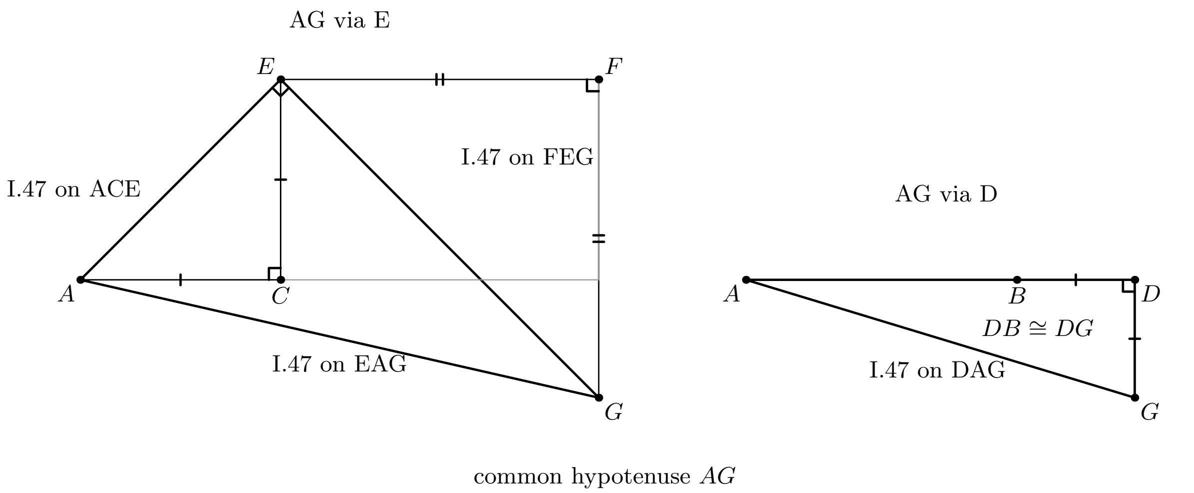

Figure 5: The four I.47 calls form two geometric paths to the same side AG. The left panel shows the E-path: I.47 on EAG reduces the square on AG to the squares on AE and EG, while the right triangles ACE and FEG expand those two leg squares. The right panel shows the D-path: I.47 on DAG reduces the same square on AG to the squares on AD and DG. The already proved segment congruences later transport DG to DB, the two $A/C/E$ legs to AC, and the two $F/E/G$ legs to CD. No literal square representatives are drawn: the formal proof may choose them independently and then identifies their content by square transport.

Diagrammatic audit. There is no further figure for Step 8 of the formal proof. After Figure 5, all spatial information needed by the argument is already visible. The remaining operations identify independent square representatives and compose equicomplementability relations; they are content bookkeeping rather than a new Euclidean state.

The classical local dependency picture is therefore

$$\begin{gathered}\text{I.11} + \text{I.3: perpendicular at } C \text{ and copied half}\\ \Downarrow \\ \text{I.31: parallels } EF \parallel AD,\ DF \parallel CE \\ \Downarrow \\ \text{Post.5: produced } EB \text{ and } FD \text{ meet at } G \\ \Downarrow \\ \text{I.5} + \text{I.15} + \text{I.29} + \text{I.32} + \text{I.6: } DB = DG,\ GF = EF \\ \Downarrow \\ \text{I.34: } EF = CD \\ \Downarrow \\ \text{four applications of I.47} \\ \Downarrow \\ \text{II.10.}\end{gathered}$$

4 The Geometric Identity in the Current Formalization

The public theorem assumes

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hABD : Geo.Between A B D)
```

The midpoint package supplies $A - C - B$ and $AC \cong CB$. The second hypothesis places D beyond B. The proof then derives, among other order facts,

$$A - C - B - D, \qquad C - B - D, \qquad A - C - D, \qquad D - B - A.$$

The theorem constructs concrete squares on AD, DB, AC, and CD. If their formal terms are denoted by

$$S_{AD}, \quad S_{DB}, \quad S_{AC}, \quad S_{CD},$$

the conclusion is

$$S_{AD} + S_{DB} \simeq_{\text{ec}} (S_{AC} + S_{AC}) + (S_{CD} + S_{CD}),$$

where $\simeq_{\text{ec}}$ denotes `HilbertScissorsEquicomplementable`.

This conclusion is not `HilbertScissorsEq`. The proof does not claim that one explicit finite dissection of the two left-hand target squares is the four right-hand target squares. Rather, the current I.47 theorem supplies equicomplementability, and II.10 composes four such Pythagorean blocks with square-representative transport.

5 Reusable Infrastructure Used

5.1 A generic perpendicular-and-copy construction

The helper

```
proposition2_9_erect_perpendicular
```

is defined in `Proposition2_9Construction.lean`, but its statement is not specific to II.9. It packages the source operations I.11 and I.3: given a nondegenerate segment, it produces a point off its carrier so that the new segment is perpendicular to the old one and congruent to a prescribed segment.

II.10 reuses this helper to construct E with $CE \perp AC$ and $CE \cong AC$. The midpoint congruence then gives $CE \cong CB$.

5.2 Parallelogram geometry

The formal proof constructs the fourth vertex F of the parallelogram $CDFE$ through the established Euclidean parallelogram interface. This packages the parallel-existence geometry corresponding to Euclid's I.31. The resulting `IsParallelogram` object supplies the two parallel pairs, while Proposition I.34 supplies opposite-side congruences such as

$$CD \cong FE.$$

Right-angle propagation through the parallelogram uses the reusable theorem `parallelogram_adjacent_right_angle`. This is the formal interface form of the I.29 parallel-angle mechanism needed in the source proof.

5.3 Angle decomposition without numerical measure

Euclid repeatedly reasons with "half a right angle" and subtracts equal angle components from equal right angles. The Lean proof introduces no degrees or real-valued angle measure. It uses explicit `HilbertRayMeetsSegment` witnesses together with

```
hilbert_angleDecomposition_halves_congruent_of_whole_congruent
hilbert_angleDecomposition_angle_addition_interior
hilbert_angleDecomposition_angle_subtraction
```

from `HilbertAngleDecomposition.lean`.

The exterior-angle form of I.32 supplies the points used to witness these decompositions. This is the formal replacement for Euclid's compressed angle arithmetic.

5.4 Directed intersection and plane separation

The source says that EB and FD, when produced in the directions B and D, meet by Postulate 5. The formal proof must recover more than the bare existence of an intersection: it needs

$$E - B - G, \qquad F - D - G.$$

The helper `proposition2_10_oriented_intersection` proves this directed form. It uses Euclidean parallel uniqueness together with same-side/opposite-side information, segment-line intersection, line uniqueness, and order trichotomy. This is a characteristic example of hidden formal data that the classical drawing suppresses.

5.5 Book I square construction, Pythagoras, and square transport

The proof invokes `euclid_proposition_47` four times and later uses `euclid_proposition_46` to choose the four square representatives appearing in the public theorem. Since independent I.47 calls may choose different squares on the same or congruent side, the reusable theorem `hilbert_square_transport` identifies their content.

This is representation transport. It is not an equation in a numerical ring and does not introduce a primitive square operation on segment lengths.

6 Proof Architecture of the Reconstruction

6.1 Step 1: normalize the external order

`proposition2_10_order` starts from the midpoint and extension hypotheses and establishes the order package needed later:

$$A - C - B, \qquad C - B - D, \qquad A - C - D, \qquad D - B - A.$$

The source statement only draws these positions; the formal proof records them individually because later ray transports depend on orientation.

6.2 Step 2: erect the perpendicular and copy the half

`proposition2_10_configuration` constructs E and proves

$$CE \cong AC, \qquad CE \cong CB,$$

together with noncollinearity and the right angles at C required for the two isosceles triangles ACE and ECB.

6.3 Step 3: build the auxiliary parallelogram

`proposition2_10_parallelogram` constructs F so that

$$CDFE$$

is a parallelogram. The helper returns the parallel pairs, the opposite-side congruences

$$CD \cong FE, \qquad DF \cong EC,$$

and the right angle at F that will later enter triangle FEG.

6.4 Step 4: reconstruct Euclid's first angle block

`proposition2_10_classical_angle_block` applies I.5 in triangles ACE and CEB and uses I.32 to create explicit angle-decomposition witnesses. It proves

$$\angle AEC \cong \angle BEC$$

and hence establishes the right angle AEB.

The formal witnesses R and S lie on the appropriate produced sides and encode the two half-right-angle decompositions. They are construction data, not part of the final theorem statement.

6.5 Step 5: produce the intersection in the correct direction

`proposition2_10_oriented_intersection` produces G and proves the strict source orders

$$E - B - G, \qquad F - D - G.$$

The proof does not accept an arbitrary intersection and then reason from the picture. It proves that the intersection lies on the required extensions.

6.6 Step 6: recover the two isosceles equalities

`proposition2_10_classical_isosceles_blocks` follows the source angle route to obtain

$$DB \cong DG, \qquad GF \cong EF.$$

At B, I.15 supplies the vertical-angle bridge; at D the parallel configuration gives a right angle; I.32 identifies the remaining half-right angle, and I.6 yields $DB \cong DG$.

The second block transports the corresponding half-right angle to triangle FEG, uses the right angle at F, and again finishes with I.6. Local interior-ray witnesses introduced by I.32 make Euclid's subtraction of angle parts explicit.

6.7 Step 7: package the four Pythagorean blocks

`proposition2_10_four_pythagoras_blocks` establishes the segment identifications

$$CA \cong CE, \qquad FE \cong CD, \qquad GF \cong CD, \qquad DB \cong DG$$

and applies I.47 to exactly four right triangles:

$$ACE, \qquad FEG, \qquad EAG, \qquad DAG.$$

The helper returns the four existential square packages and their `HilbertScissorsEquicomplementable` decompositions.

6.8 Step 8: follow the common square on AG

The final theorem uses I.46 to choose statement-level squares $S_{AD}, S_{DB}, S_{AC}, S_{CD}$. It begins from I.47 on DAG:

$$S_{AG} \simeq_{\mathrm{ec}} S_{DA} + S_{DG}.$$

Square transport replaces DA by AD and DG by DB, yielding the left-hand target.

A second representative of the square on AG, coming from triangle EAG, is then identified with the first. I.47 on EAG gives

$$S_{AD} + S_{DB} \simeq_{\text{ec}} S_{EA} + S_{EG}.$$

The remaining I.47 blocks expand these two squares:

$$S_{EA} + S_{EG} \simeq_{\text{ec}} (S_{CA} + S_{CE}) + (S_{FE} + S_{FG}).$$

Finally

$$CA \cong AC, \quad CE \cong AC, \quad FE \cong CD, \quad FG \cong CD$$

transport the four terms to two copies of S_{AC} and two copies of S_{CD}. Additive compatibility and transitivity close the proof. No content cancellation is used.

7 Lean Formalization

7.1 The public theorem signature

The following is an ASCII transcription of the current compiling source:

```
theorem euclid_proposition_2_10_oriented
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hABD : Geo.Between A B D) :
    exists SAD0 SAD1 SDB0 SDB1 SAC0 SAC1 SCD0 SCD1 : Geo.Point,
      IsSquare Geo A D SAD0 SAD1 /\
      IsSquare Geo D B SDB0 SDB1 /\
      IsSquare Geo A C SAC0 SAC1 /\
      IsSquare Geo C D SCD0 SCD1 /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A D SAD0 SAD1 +
         hilbertParallelogramTerm Geo D B SDB0 SDB1)
        ((hilbertParallelogramTerm Geo A C SAC0 SAC1 +
          hilbertParallelogramTerm Geo A C SAC0 SAC1) +
         (hilbertParallelogramTerm Geo C D SCD0 SCD1 +
          hilbertParallelogramTerm Geo C D SCD0 SCD1)) := by
```

The suffix `_oriented` is substantive. The theorem fixes the external branch $A - C - B - D$ through the hypothesis `hABD : Geo.Between A B D`. A future theorem that packages both orientations would be a separate wrapper.

7.2 The proposition-local theorem chain

The production source is factored into

```
proposition2_10_order
proposition2_10_configuration
proposition2_10_parallelogram
proposition2_10_classical_angle_block
```

```
proposition2_10_oriented_intersection
proposition2_10_classical_isosceles_blocks
proposition2_10_four_pythagoras_blocks
euclid_proposition_2_10_oriented
```

This chain mirrors genuine mathematical stages rather than merely splitting a large tactic script. The hidden witnesses E, F, G, R, S and the local I.32 decomposition points are progressively introduced and then eliminated from the final public interface.

7.3 The final scissors bridge

The last stage uses only forward equicomplementability operations:

(a) I.47 on DAG connects a square on AG with the two left target squares after congruence transport;

(b) two independently constructed AG squares are identified;

(c) I.47 on EAG replaces AG by the EA and EG squares;

(d) I.47 on ACE and FEG expands those into four leg squares;

(e) square transport identifies the four legs with two copies of AC and two copies of CD.

No subtraction or cancellation in the scissors calculus is used.

8 Relationship with Earlier Book II Propositions

The production theorem does not call an earlier Book II proposition. In particular it does not use II.9 as an identity from which II.10 is deduced.

There is one important module-level nuance. The import `Proposition2_9Construction` supplies `proposition2_9_erect_perpendicular`, a generic construction helper encoding I.11 plus I.3. Thus II.10 reuses infrastructure that was extracted during the II.9 development, but it does not reuse the theorem `euclid_proposition_2_9_oriented` or any II.9 content identity.

This distinction is mathematically appropriate. Euclid's guide-level relationship between II.9 and II.10 is that the diagrams and arguments are companions: II.10 moves the unequal point beyond the end of the original segment. The formal library preserves that parallelism by sharing low-level construction machinery while proving the two proposition-level statements separately.

9 Relationship with Book I Infrastructure

The classical source uses I.11 and I.3 to erect and copy the perpendicular, I.31 for the parallels, I.29 and Postulate 5 for the produced intersection, I.5 for the isosceles base angles, I.15 for the vertical angle, I.32 for the half-right-angle arguments, I.6 for the two isosceles conclusions, I.34 for opposite sides of the parallelogram, and I.47 four times [**?**].

The Lean DAG packages several of these steps. The I.11+I.3 construction is the reusable perpendicular helper. I.31 is represented by the fourth-vertex/parallelogram interface. The I.29

right-angle consequences are available through parallelogram right-angle transport. Postulate 5 is replaced at the proof-script level by a Euclidean parallel-uniqueness contradiction together with plane-separation and order machinery that also determines the direction of the intersection.

Thus the formal proof is source-faithful in mathematical architecture but is not a line-by-line transcription of Euclid's prose.

10 Formalization Notes and Hidden Geometric Data

10.1 The external order is a theorem input

II.10 is not the internal-cut branch of II.9 with names changed. The hypothesis

```
hABD : Geo.Between A B D
```

places D strictly beyond B. From this the proof derives the other betweenness relations required to compare rays from C and D.

10.2 The intersection G needs orientation, not just existence

The classical phrase "produce the lines until they meet" hides a major formal obligation. Later angle identities require B to lie between E and G and D to lie between F and G. An arbitrary point lying on both carriers is insufficient.

The formal proof therefore combines side-of-line data, order trichotomy, and line uniqueness to establish the exact orders $E - B - G$ and $F - D - G$.

10.3 Same-ray witnesses control angle-arm replacement

Once the strict orders are known, `HilbertSameRay` witnesses justify replacing EB by EG, DB by DA or DC where appropriate, and FD by FG. These transports are invisible in the printed diagram but essential to every right-angle propagation.

10.4 The points R, S and later local witnesses encode angle arithmetic

The I.32 exterior-angle theorem produces concrete points that witness interior rays. The first pair supports the proof that AEB is right. Further local witnesses support the two synthetic angle-subtraction arguments leading to $DB = DG$ and $GF = EF$.

No statement of the form "$45° + 45° = 90°$" occurs in the Lean development. The information is carried by betweenness, ray intersection, and angle congruence.

10.5 Noncollinearity is structural data

Every use of right-angle transport, I.6, I.15, angle decomposition, and I.47 requires nondegeneracy. The proof repeatedly derives noncollinearity from strict betweenness, the perpendicular construction, and the parallelogram. These are genuine geometric facts, not merely type-checking noise.

10.6 Square representatives are not canonical

Four I.47 calls and four final I.46 constructions create several squares on the same or congruent segments. The proof must therefore identify their content explicitly with `hilbert_square_transport`. The final identity is independent of these representation choices, but the proof of that independence is part of the formal content layer.

11 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. Its semantic layers are

$$\begin{gathered}\text{incidence/order/congruence and Euclidean parallel geometry}\\ \Downarrow\\ \text{perpendiculars, right angles, angle decomposition, parallelogram facts}\\ \Downarrow\\ \text{concrete square representatives}\\ \Downarrow\\ \text{I.47 equicomplementability} + \text{square transport}\\ \Downarrow\\ \text{II.10 equicomplementability conclusion.}\end{gathered}$$

The transitive axiom audit was run on the compiled production theorem:

```
#print axioms Geometry.euclid_proposition_2_10_oriented
```

and reported

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise conclusion is therefore: II.10 adds no new axiom and no `sorry`; it is not literally axiom-free.

The theorem uses neither numerical area nor the later arithmetic of segment multiplication. The algebraic-looking identity remains entirely in the synthetic geometry/content architecture.

12 Dependency Summary

The production chain is

$$\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer}\\
\Downarrow\\
\text{I.11 + I.3 generic perpendicular/copy infrastructure}\\
\Downarrow\\
\text{order + perpendicular configuration + parallelogram } CDFE\\
\Downarrow\\
\text{I.5 + I.15 + I.29/I.32 + angle-decomposition machinery}\\
\Downarrow\\
\text{oriented intersection } E-B-G,\ F-D-G\\
\Downarrow\\
\text{I.6: } DB=DG,\ GF=EF \quad + \quad \text{I.34: } EF=CD\\
\Downarrow\\
\text{four I.47 blocks on } ACE, FEG, EAG, DAG\\
\Downarrow\\
\text{I.46 target squares + HilbertSquareTransport}\\
\Downarrow\\
\text{equicomplementability addition/transitivity}\\
\Downarrow\\
\texttt{euclid_proposition_2_10_oriented}.
\end{gathered}$$

The implementation checkpoint is:

New geometric axiom in II.10:	no
New content axiom in II.10:	no
New proposition-local helper theorem:	yes
New reusable public helper theorem/module:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly calls an earlier Book II proposition:	no
Imports II.9 production theorem:	no
Reuses generic helper from `Proposition2_9Construction`:	yes
Conclusion is exact `HilbertScissorsEq`:	no
Conclusion is `HilbertScissorsEquicomplementable`:	yes
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `#print axioms`:	`propext`, `Classical.choice`, `bookZero_nullSegment1`, `Quot.sound`

13 Conclusion

Proposition II.10 confirms that the architectural change first visible in II.9 is a genuine new Book II proof mode rather than a one-off exception. The rectangle-normal-form chain of II.1–II.8 is not used. Instead the formalization follows Euclid's external-section construction through perpendiculars, parallels, produced intersections, isosceles angle arguments, and four Pythagorean content identities.

The most delicate new formal feature is the direction of the intersection G. The source drawing makes $E - B - G$ and $F - D - G$ immediate; Lean must prove both. Once that order is available, the remaining geometry follows the classical metric bridges $DB = DG$ and $GF = EF = CD$.

The final common object is the square on AG. One I.47 path connects it to the two target squares on AD and DB; the other connects it to the four leg squares that become two copies of AC and two copies of CD. Consequently the mnemonic

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

is best understood as the algebraic shadow of a synthetic equicomplementability theorem. The current proof does not use II.9 as a proposition, numerical area, or segment multiplication, and the completed axiom audit shows that II.10 introduces no new axiom beyond the established transitive project dependencies.